consequences of the two twentieth-century revolutions in physics: Einstein's general relativity, with its new viewpoints on space and time, was showing up in the gravitational redshift of light from Sirius B; and the new quantum mechanics, with its wave/particle duality, was responsible for Sirius B's internal pressure. This astronomy was a fertile field in which a young man could make his mark.

As he continued his university studies in Madras, Chandrasekhar explored further the consequences of quantum mechanics for the astronomical Universe. He even wrote a small article on his ideas, mailed it to England to R. H. Fowler, whom he had never met, and Fowler arranged for it to be published.

Finally, in 1930 at age nineteen, Chandrasekhar completed the Indian equivalent of an American bachelor's degree, and in the last week of July he boarded a steamer bound for far-off England. He had been accepted for graduate study at Cambridge University, the home of his heroes, R. H. Fowler and Arthur Eddington.

The Maximum Mass

Those eighteen days at sea, steaming from Madras to Southampton, were Chandrasekhar's first opportunity in many months to think quietly about physics without the distraction of formal studies and examinations. The solitude of the sea was conducive to thought, and Chandrasekhar's thoughts were fertile. So fertile, in fact, that they would help to win him the Nobel Prize, but only fifty-four years later, and only after a great struggle to get them accepted by the world's astronomical community.

Aboard the steamer, Chandrasekhar let his mind reminisce over white dwarfs, Eddington's paradox, and Fowler's resolution. Fowler's resolution almost certainly had to be correct; there was none other in sight. However, Fowler had not worked out the full details of the balance between degeneracy pressure and gravity in a white-dwarf star, nor had he computed the star's resulting internal structure—the manner in which its density, pressure, and gravity change as one goes from its surface down to its center. Here was an interesting challenge to help ward off boredom during the long voyage.